IS1203: Foundations of Information Systems

Topic 2:

Information Systems in Organizations

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Recommended Text:
https://paginas.fe.up.pt/~acbrito/laudon/index.htm
http://wps.prenhall.com/bp_laudon_essmis_6/21/5555/1422269.cw/-/t/index.html

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Intended Learning Outcomes

After completion of this topic, you'll be able to,

- Analyze the relationship between organizations and information technology
- Describe the role of information systems within the value-added processes of an organization
- Discuss how organizational structure, culture, and change affect the implementation of information systems
- Describe how organizations should use information systems to gain a competitive advantage
- Describe how information systems can be used to address organizational change

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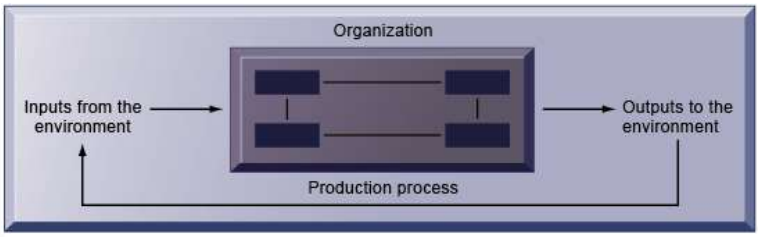
Organization

- **Organization** is a formal collection of resources and people that is structured and managed to accomplish a set of goals.
- It is **an open system** - It affects and is affected by its surrounding environment
- From **a technical view**, an organization is **a formal, legal, social structure** that processes resources, or inputs, to produce outputs.



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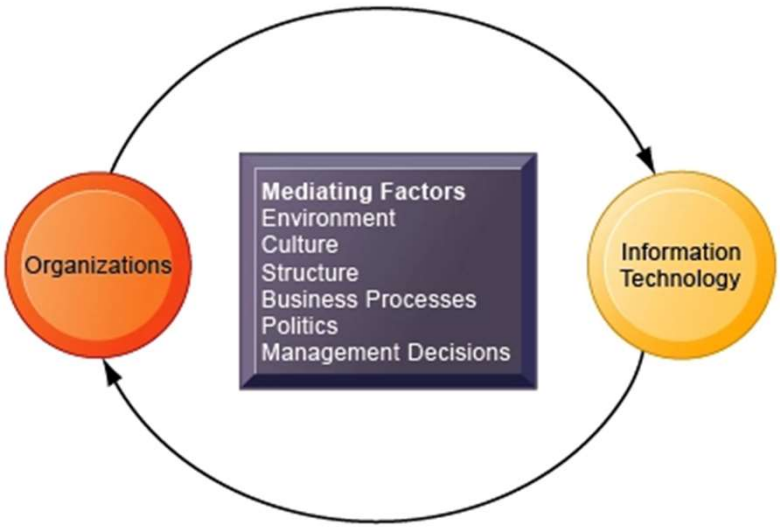
Relationship Between IT and Organization

- There's **a complex two-way relationship** mediated by many factors
- **To design new systems or improve existing systems** successfully, first you should **understand** the business organization well.



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Exercise:

1. Identify different stakeholders of an organization.
2. Identify the members of the organization and the relationships between them and the other stakeholders.
3. List different processes and activities they perform.
4. Are the organizational processes automated using well-defined information systems?

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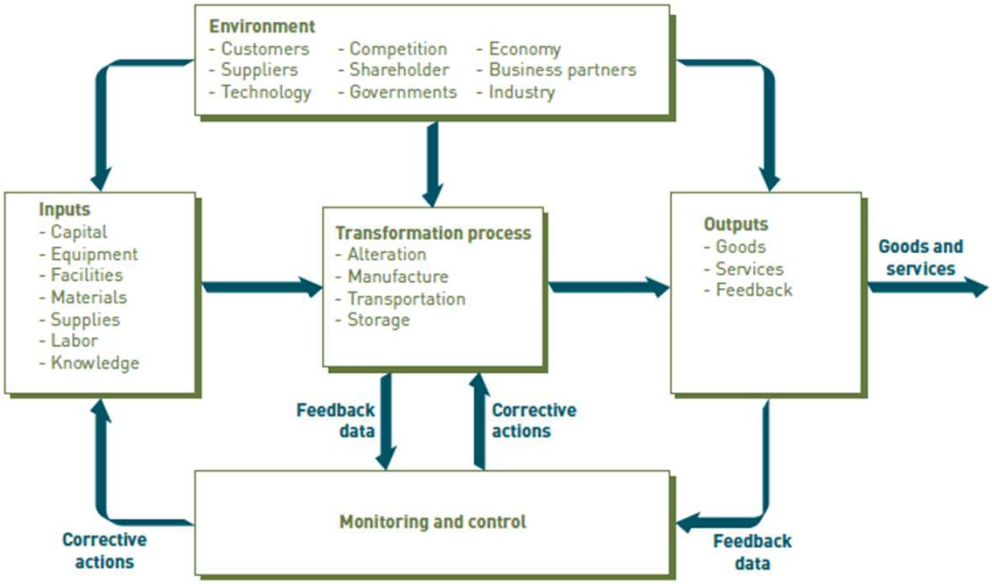
A General Model of an Organization



[From the recommended Text- Fundamentals of Information Systems]

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Functions, Processes and Activities of an Organization

- **Management and Leadership**
  - Strategic planning> Preparing vision and mission, setting goals and objectives, Strategy formulation and implementation, …
  - Organizing
  - Change management
  - Innovation and improvement
  - Leading and controlling
- **Human Resources**
- **Finance and Accounting**
- **Marketing and Sales**
- **IT and Cooperate Communications**
- **Research and Development**
- **Legal and Compliance**
- **Customer Service**

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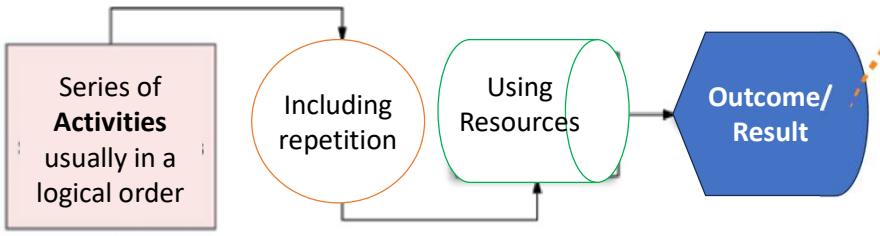
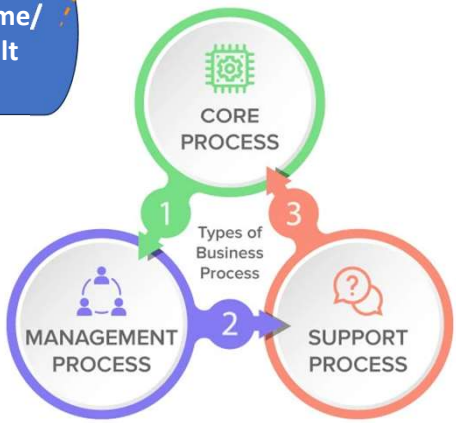
Business Process



Adapted from (Riutta, 2019)



A good process meets three essential criteria:

- **Repeatability**
- **Transparency**
- **Agility**

(NotifyVisitors, 2022)

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Value Adding

It is the work that

- the customer is willing to pay for and
- physically transforms the product/service or
- is done right the first time.

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Value Chain of an Organization

- A series (chain) of activities and processes involved in creating a product or performing a service.
- The value that's created and captured by a company is considered for calculating the profit:

Gross Profit = Value Created & Captured – Cost of Creating that Value

- The more value an organization creates, the more likely it is to be profitable.

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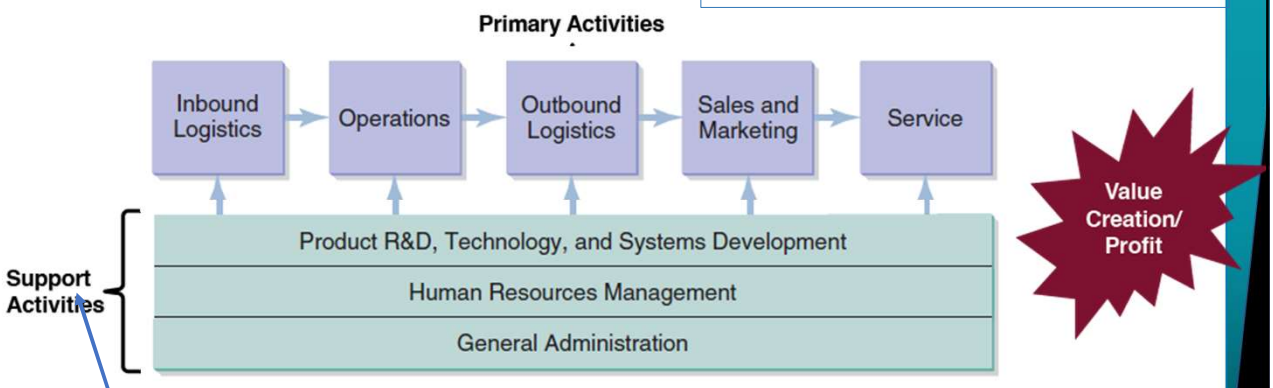
Value Chain of an Organization…contd.

Activities that are directly related to the production and distribution of the products and services



To make the delivery of the primary activities possible

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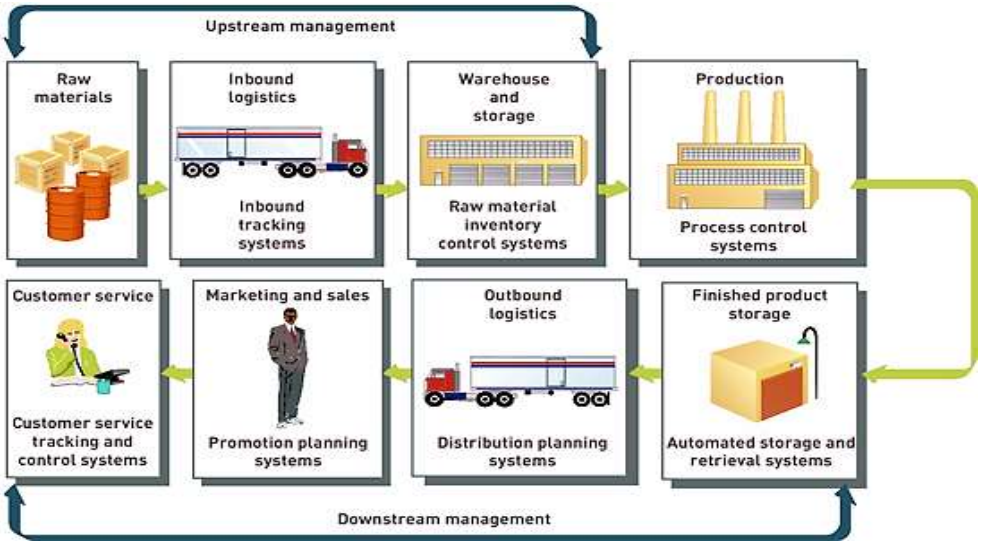
e.g. Value Chain of a Manufacturing Company



Source: The Value Chain of a Manufacturing Company. Fundamentals of Information Systems

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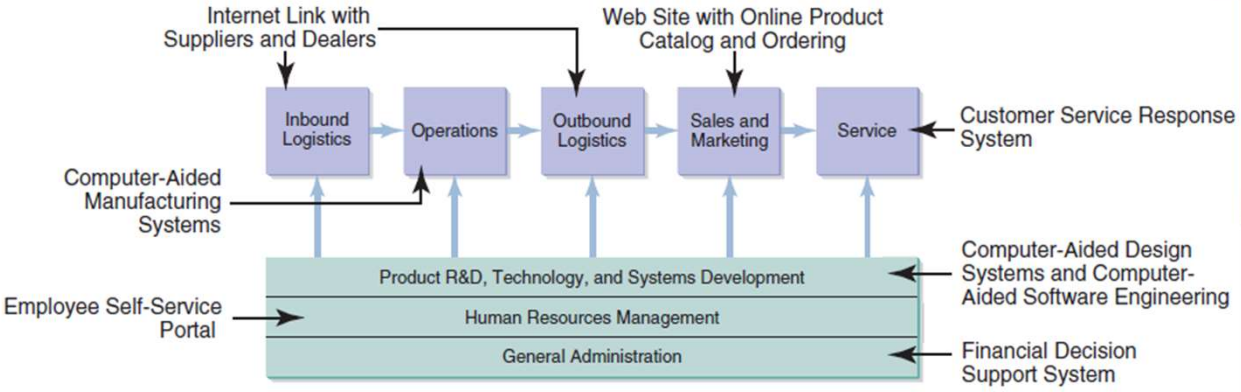
How ISs can be used to add value



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Want to Introduce an IS?

- Study the Organization First

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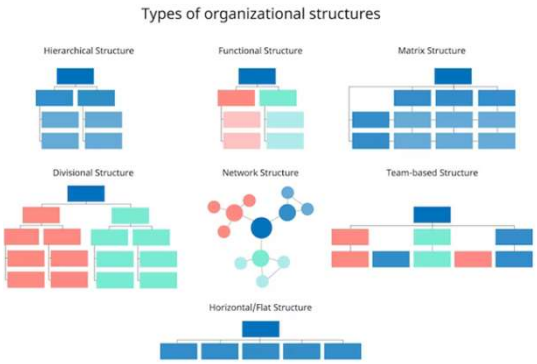
Organizational Structures

- **Organizational structure:** organizational subunits and the way they relate to the overall organization

- Categories of organizational structure
  (1) Traditional
  (2) Project/Team
  (3) Matrix
  (4) Virtual



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(1)Traditional Organizational Structure

- A hierarchical structure
- Major department heads report to a president or top-level manager
- Managerial pyramid shows the hierarchy of decision-making and authority

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(1) Traditional Organizational Structure cont….

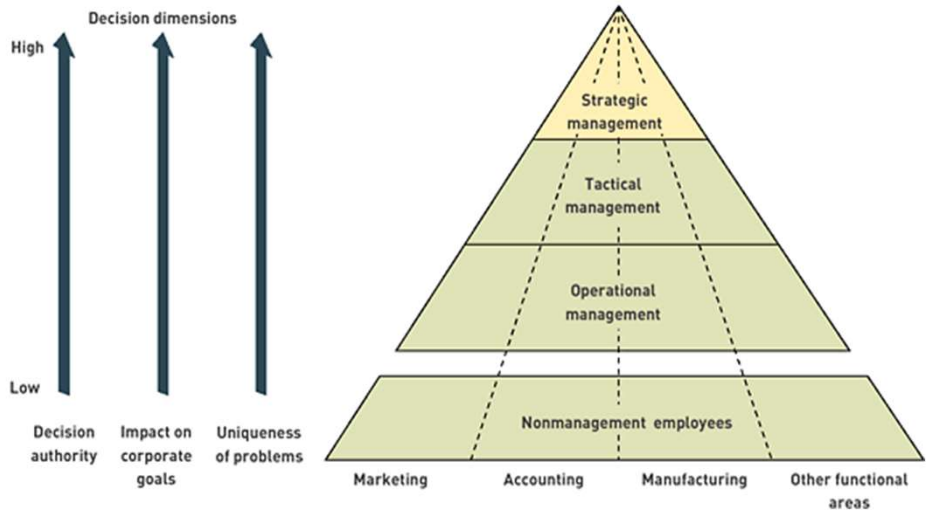

Figure : A simplified model of the organization, showing the managerial pyramid from top-level managers to non management employees

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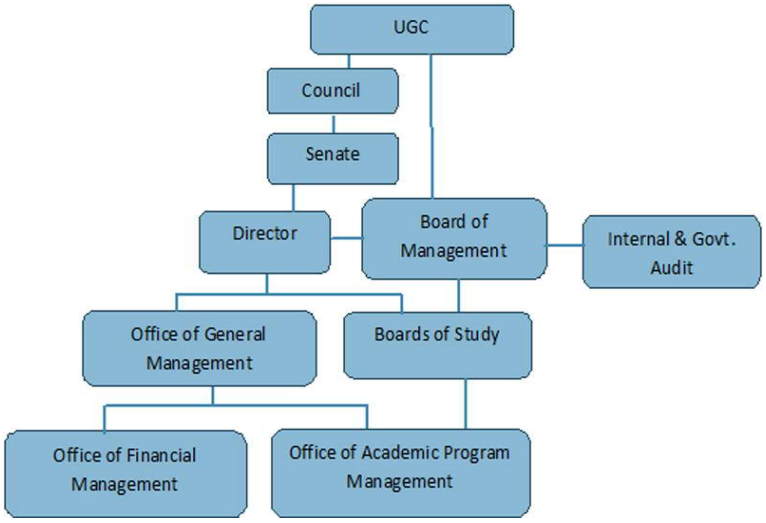
(1) Traditional Organizational Structure cont..

Example:



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(2) Project/Team Organizational Structures

- **Project organizational structure:** Centred on major products or services
  - Many project teams are often formed temporarily
- **Team organizational structure:** Centred on work teams or groups
  - Team can be temporary or permanent, depending on the tasks

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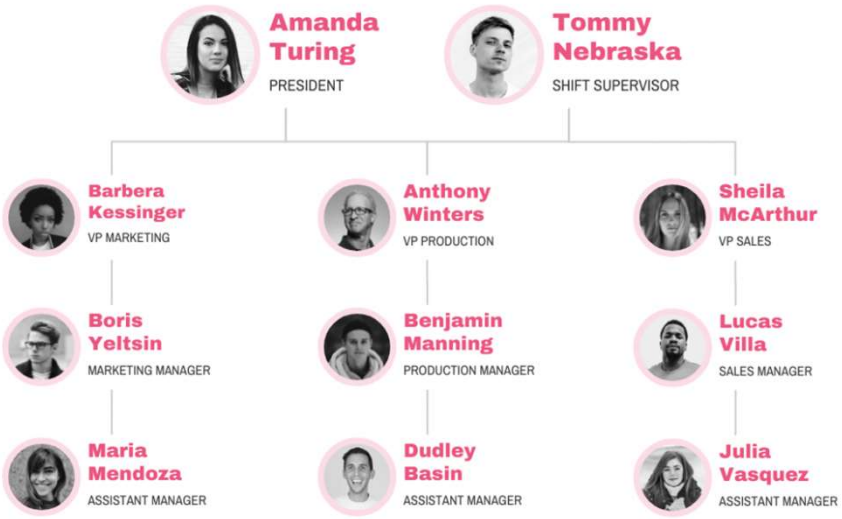
(2) Project/Team Organizational Structures cont….



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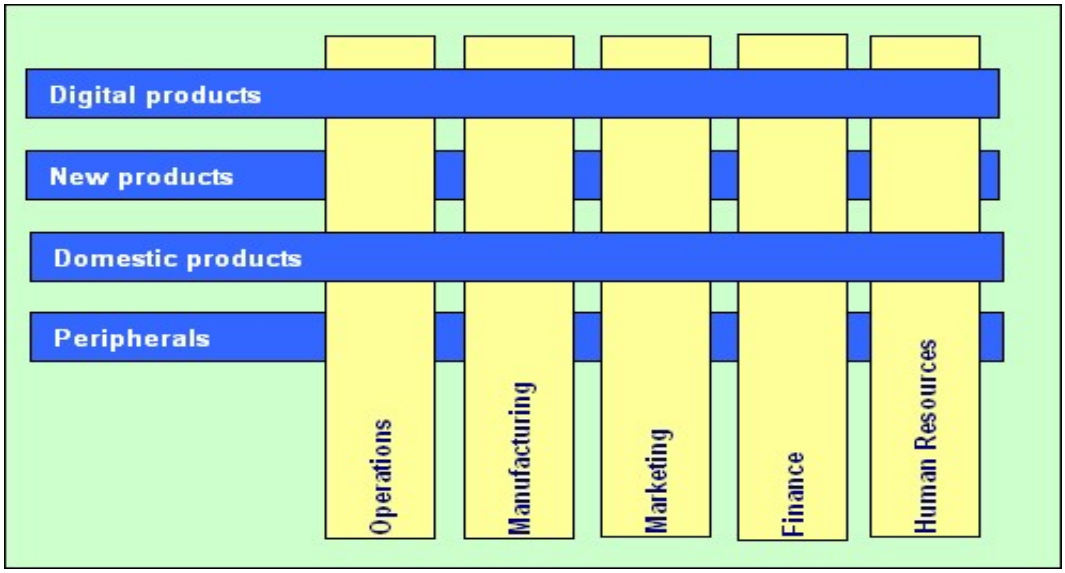
(3) Matrix Structure



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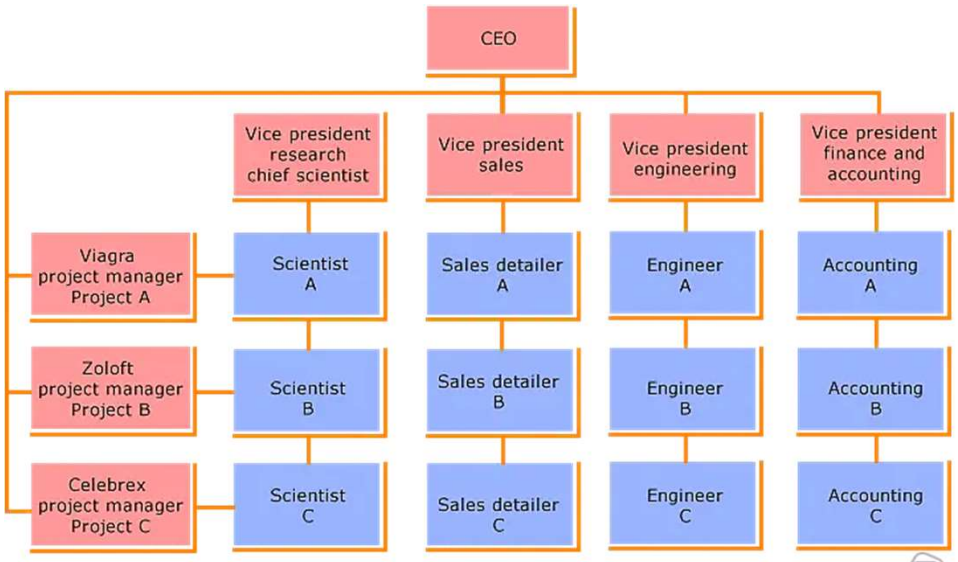
e.g. Matrix Organizational Structure



Possible Matrix Organizational Structure of Pfizer Corporation| Source: *Introduction to Business*, by Julian Gaspar, Leonard Bierman, et al.

ifioque.com

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(4) Virtual Organizational Structure and Collaborative Work

- Virtual organizational structure
  - Employs business units in geographically dispersed areas
  - People may never meet face to face
  - Allows collaborative work
    - Managers and employees can effectively work in groups, even those composed of members from around the world

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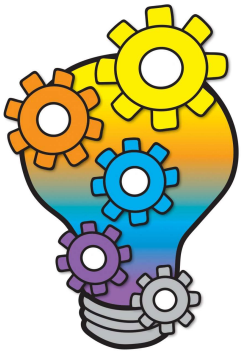
Key Elements of an Organizational Structure



- Chain of command
- Roles and responsibilities
- Scope of control
- Decision-making authority
- Departments or teams within the organization

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Organizational Values

- **Integrity:** Commitment to honesty, transparency, and ethical behaviour.
- **Striving for excellence:** Aim for the highest quality and continuous improvement.
- **Accountability:** Taking responsibility for actions, decisions, and their outcomes.
- **Encourage Innovations**
- **Customer Focus**
- **Respecting each other**
- **Teamwork**
- **Sustainability**
- **Community Concerned**
- **Employee Empowerment**

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Exercise

1. What is the importance of studying the organizational structure when identifying information system requirements?

2. Suppose that SL Police is finding it difficult to track the records of criminals and identify them easily. You have been requested to propose a solution to this issue.
   a) List the activities you should do.
   b) Justify the importance of each activity you listed for question (a).

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Organizational Culture

- Is the "personality" of an organization.
- Defined as the shared values, beliefs, norms and expectations that govern the way people approach their work and interact with each other.
- Contains the unwritten rules and social practices that shape the work environment and guide how employees conduct themselves in their roles.
- Influences the usage of IS and IS can influence it.
  - **How?** Define core values, Lead by example, Foster open communication, Recognise and reward, Promote collaboration and inclusion and Monitor the changes.
- By understanding, shaping, and nurturing organizational culture, we can create a work environment that drives overall success.

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Organizational Change

- Relates to the process of transforming an organization's strategies, structure, processes (operational methods), technologies and culture
- Can be driven by internal or external factors

1. Internal Factors: Organizational growth, Performance issues, Desire to foster innovations, and Leadership changes:
2. External Factors: Market Dynamics such as customer preferences, Technological Advances, Regulatory Changes, and Globalization

- For an effective organizational change, it is important to have a clear vision, objective, strong leadership, effective communication, employee involvement, training support, and a mechanism to monitor and adapt based on lessons learnt.

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Identifying Requirements For Change

1. Study the existing business process and its issues
2. Identify internal and external factors driving change

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Business Process Analysis

1. Plan for the analysis (Define the objectives and the scope)
2. Gather data (from stakeholders via interviews/surveys)
3. Map the current processes (Draw flowcharts and identity all activities, inputs and outputs)
4. Analyze the current process (Identify issues/bottlenecks and their causes)
5. Design improvements (Propose improvements/redesigns and assess their feasibility and impacts)
6. Implement changes (Plan, Communicate and Execute)
7. Monitor and Review

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Common Issues in Business Processes

1. Inefficiencies (Slowed down or halted processes, Redundancies, complicated procedures)
2. Poor communication (Lack of information sharing, Miscommunication or Inconsistent channels)
3. Resource constraints (Insufficient/underutilised resources and Unbalanced Workloads)
4. Lack of standardisation (Inconsistent Processes or non-compliance, No best practices)
5. Inadequate technology (Outdated Systems, Integration Issues and lack of automation)
6. Quality issues (High error rates, Reworks and Customer complaints)
7. Lack of visibility and control (Poor Monitoring, Limited Data Access and Unclear Accountability)
8. Cultural resistance (Change Aversion and Lack of support from key stakeholders)
9. Inadequate planning and design (Poor process design, risk management and strategic alignment)
10. Customer experience issues (Long cycle times, Inflexible processes, Poor quality)

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Solutions: Business Process Reengineering and Continuous Improvement

- Business Process Reengineering
  - Also called business process redesign
  - It is a radical redesign of business processes, organizational structures, information systems, and values of the organization to achieve a breakthrough in business results

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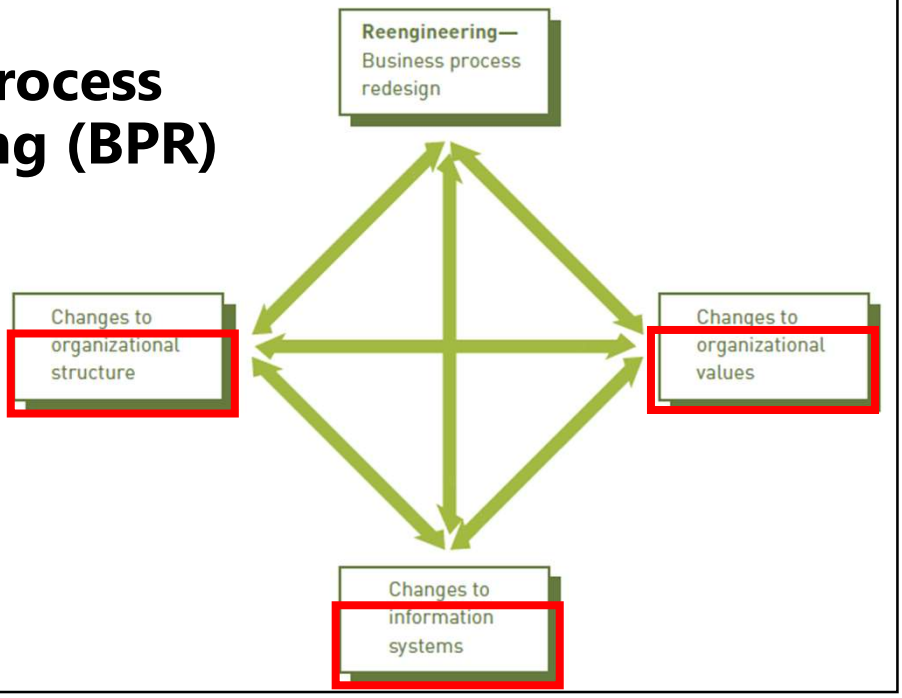
Business Process Reengineering (BPR)



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Reengineering and Continuous Improvement (continued)

- **Continuous improvement**
  - Constantly seeking ways to improve business processes
  - Benefits:
    - Increased customer loyalty
    - Reduction in customer dissatisfaction

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Exercise:

- Compare business process reengineering (BPR) and continuous improvement (BPCI).

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Comparing BPR and CI

Business Process Reengineering	Continuous Improvement
Strong action taken to solve serious problem	Routine action taken to make minor improvements
Top-down change driven by senior executives	Bottom-up change driven by workers
Broad in scope; cuts across departments	Narrow in scope; focuses on tasks in a given area
Goal is to achieve a major breakthrough	Goal is continuous, gradual improvements
Often led by resources from outside the company	Usually led by workers close to the business
Information systems are integral to the solution	Information systems provide data to guide the improvement team

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Common Changes: Outsourcing, On-Demand Computing, and Downsizing

- **Outsourcing:** contracting with outside professional services
- **On-demand computing:** contracting for computer resources to rapidly respond to an organization's varying workflow
  - On-demand business
  - Utility computing
- **Downsizing:** reducing number of employees

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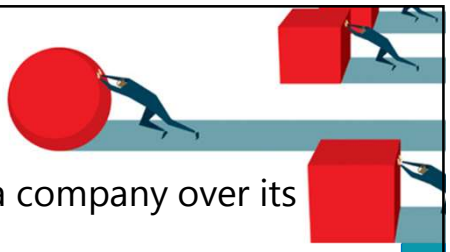
Competitive Advantage



- Significant and (ideally) long-term benefit to a company over its competition.
- An **advantage that a firm has over its competitors**, allowing it to **generate greater sales** or margins and/or **retain more customers** than its competitors.
- There can be many types of competitive advantages, including the **firm's cost structure, product offerings, distribution network and customer support**.
- Ability to establish and maintain competitive advantage is vital to the company's success

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Factors that Lead Firms to Seek Competitive Advantage

Porter's Five Forces

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Strategic Planning for Competitive Advantage

- **Cost leadership** – Reduce the cost at every level to remain competitive
- **Product/Service Differentiation** –Focus on characteristics that make your product or service stand out to your target audience
- **Niche strategy (Focused)-** Target a narrow segment of customers and meet an unmet or underserved need
- **Altering the industry structure** - Look to change the industry structure by merging with or acquiring another firm

How do organizations use IS to achieve these strategies?

Seek ways to reduce cost, and response time and improve quality, and flexibility.

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Strategic Planning for Competitive Advantage cont….

Other strategies

- First to market
- Customizing products and services
- Hiring the best people

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Competitive Advantage-Examples

Competitive Advantage	Example
Better relationship with suppliers	Internet marketplace to allow the company to purchase products from suppliers at discounted prices
Costs dramatically reduced	Warehouse management system employing RFID technology to reduce labor hours and improve inventory accuracy
Customer loyalty increased	Customer interaction system to monitor and track each customer interaction with the company
Inventory levels reduced	Collaborative planning, forecasting, and replenishing system to ensure the right amount of inventory is in stores
Superior information gathering	Order configuration system to ensure that products ordered will meet customer's objectives
Superior service provided to customers	Tracking systems that customers can access to determine shipping status

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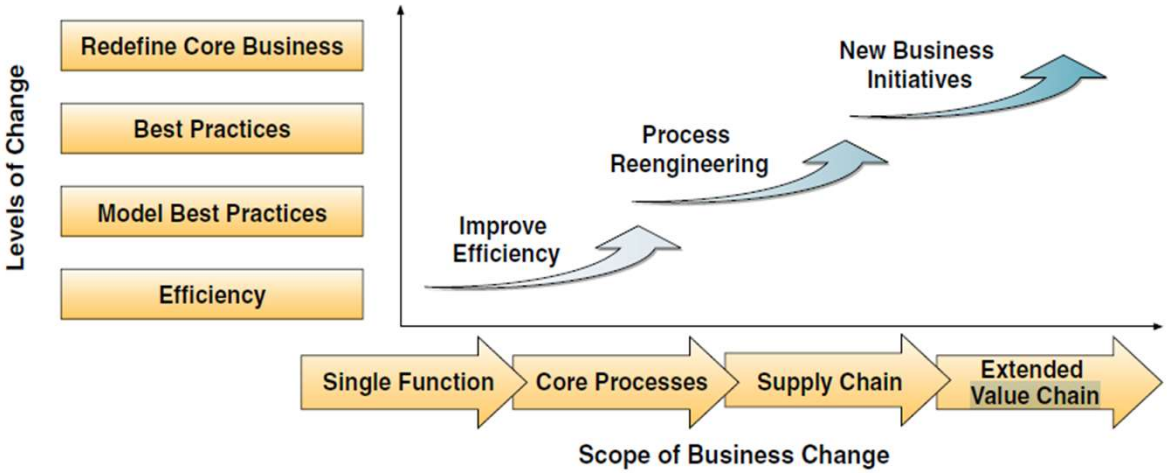
Impact and change in the levels & scope of business by an IT implementation



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Exercise

How are the following companies supported by information systems to address organizational change?

1. Agile Companies that use agile methods and principles to enhance flexibility, innovation, and rapid adaptation (e.g. Amazon and Spotify)
2. Virtual Companies that operates primarily online and offering digital products or services without relying on physical locations. (e.g. Netflix and GitHub)

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Summary:
What did we discuss on this topic?

- Relationship between organizations and Information Technology
- The role of IS within the value-added processes of an organization
- How organizational structure, culture, and change affect the implementation of information systems
- How information systems can support gaining competitive advantage
- How information systems can be used to address organizational change

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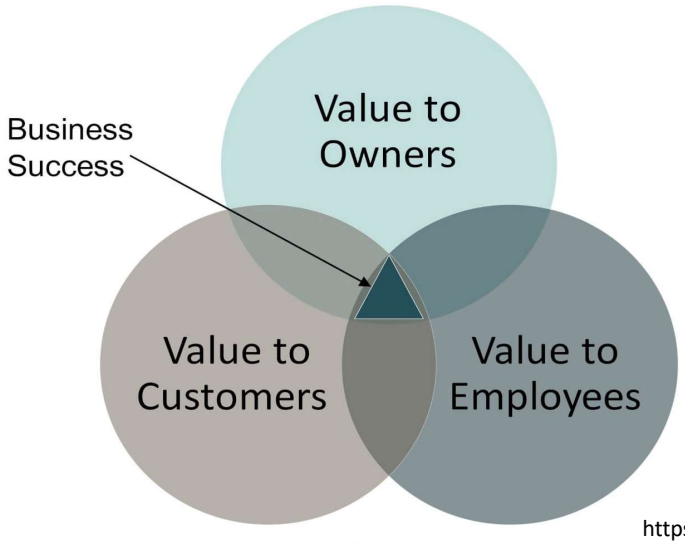
Revision: Value Creation



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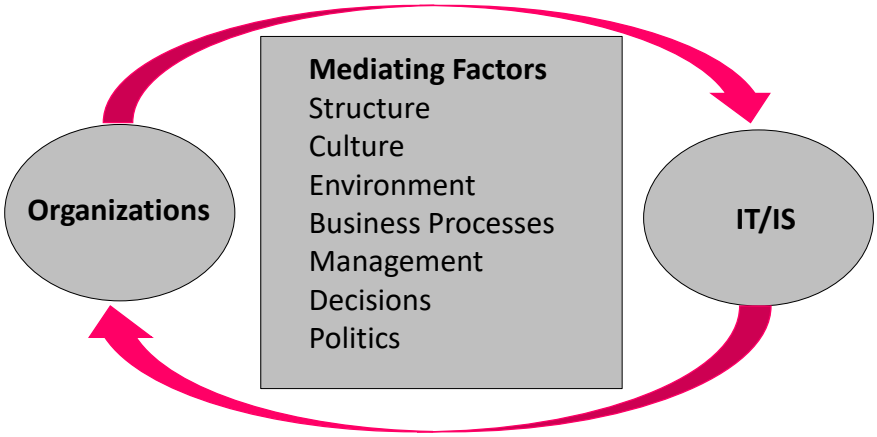
Revision: Relationship – IS and Organization



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